

Sharpening the amplitude-dyadic lift: the discarded count constraint removes half a logarithm

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

AddA v1.2's amplitude-dyadic lift made H-KBAL structurally non-load-bearing at the honest cost $64\sqrt{7}\sqrt{n}\log^2(2n)$, with the sharp-constant optimization registered as follow-up. This note delivers the first sharpening: half a logarithm, by a two-line restoration of a constraint the original proof discarded. Scope: the unconditional-amplitude theorem only; the official ledger (balanced route) and the amended-class closure are untouched.

2. The sharpened theorem

Theorem 1 (Amplitude-dyadic lift, log-sharpened). *For arbitrary positive amplitudes (no balance hypothesis),*

$$\sum_{t \neq 0} w_t^2 \leq 64\sqrt{7} \lambda'^2 I^2 \sqrt{2n} \log^{3/2}(2n) + O(\lambda'^2 I^2).$$

Proof. Identical to AddA v1.2 up to the class assembly: with $x_j := \sqrt{I_j N_j^{1/4}}$ the per-class-pair bound factorizes, $\|w^{jk}\|_{\ell_t^2} \leq \lambda' 8 \cdot 7^{1/4} x_j x_k$, so by Minkowski

$$\|w\|_{\ell_t^2} \leq \lambda' 8 \cdot 7^{1/4} \left(\sum_{j \leq J} x_j \right)^2, \quad \|w\|_{\ell_t^2}^2 \leq 64\sqrt{7} \lambda'^2 \left(\sum_{j \leq J} x_j \right)^4.$$

The original proof bounded $\sum x_j$ by Cauchy–Schwarz against the trivial $N_j \leq N$, paying J^2 inside the fourth power. Instead, keep BOTH constraints $\sum_j I_j = I$ and $\sum_j N_j = 2n$:

$$\begin{aligned} \sum_{j \leq J} x_j &= \sum_{j \leq J} I_j^{1/2} N_j^{1/8} \\ &\stackrel{\text{CS}}{\leq} \left(\sum_j I_j \right)^{1/2} \left(\sum_j N_j^{1/4} \right)^{1/2} \stackrel{\text{Jensen}}{\leq} I^{1/2} [J^{3/4} (2n)^{1/4}]^{1/2} = I^{1/2} J^{3/8} (2n)^{1/8}, \end{aligned}$$

using the concavity of $x \mapsto x^{1/4}$ (power mean: $\sum_{j=1}^J N_j^{1/4} \leq J (\sum N_j / J)^{1/4} = J^{3/4} (2n)^{1/4}$). Raising to the fourth power gives $(\sum x_j)^4 \leq I^2 J^{3/2} \sqrt{2n}$ with $J = \log_2(2n) + O(1)$; the tail classes are absorbed exactly as before. $\square$

At the amended-class scale $n = 44$: $\log_2(88) = 6.46$, $\log^2 = 41.7$ vs $\log^{3/2} = 16.4$ — a $\times 2.54$ improvement of the unconditional constant; since the route's reach scales as the inverse SQUARE of the prefactor in $K(n) \leq 8 + c\sqrt{n}$, the n -reach of the unconditional route extends by $\times 6.5$.

3. The candidate second saving (flagged, not asserted)

The per-class amplitude cap enters squared twice. With the dyadic class $A \in (2^{-j-1}a_0, 2^{-j}a_0]$ and class point count N_j : the LOWER end gives $\sum_{\text{class}} A^2 \geq N_j (2^{-j-1}a_0)^2$, i.e. $(2^{-j}a_0)^2 \leq 4(\sum_{\text{class}} A^2)/N_j$, so the cap reads $\alpha_j^2 \leq 4I_j^{\text{class}}/N_j$ where $I_j^{\text{class}} = \sum_{\text{class}} A^2$. The AddA proof uses $\alpha_j^2 \leq 8I_j/N_j$; whether the extra factor 2 is a convention difference (class intensity $2I_j$ vs I_j ; signed-mode vs reading counting) or slack is decidable only against the script's definitions. IF the 4 is available in the proof's convention, the prefactor improves $64\sqrt{7} \rightarrow 16\sqrt{7}$ (a further $\times 4$). FLAGGED as a definition-check item for the registered machine spot-check; NOT asserted here.

4. Ledger and scope impact

None on the critical path: the OFFICIAL ledger threshold remains the balanced sharp-constant route (1.59×10^5); the amended-class closure needs only $n_{\text{pack}} \approx 44$, where every route has orders of slack. The sharpening's value is (i) robustness of the unconditional-amplitude statement (its reach was the binding figure if the amendment were ever revisited), (ii) record hygiene (the honest constant should be the best simply-provable one).

5. Numerical verification

NO machine execution this session (sandbox shell unavailable). The proof's two new steps are textbook inequalities (Cauchy-Schwarz; Jensen for $x^{1/4}$); the numerical comparisons (41.7 vs 16.4 at $n = 44$; $\times 2.54$; reach $\times 6.5$) are closed-form arithmetic. The registered machine spot-check: re-run the AddA v1.2 unbalanced- profile measurements against the sharpened bound (the measured worst ratio 0.03 was against the $\log^2$ ceiling 2929; the sharpened ceiling at the same configuration is ~ 1153 , ratio ~ 0.08 — still two orders of slack) and execute the section-3 definition check.

Quantitative sanity check (mandatory): limit case — for a single occupied class ($J_{\text{occ}} = 1$, the balanced case) the sharpened assembly degenerates to $\sum x_j = x_1 = \sqrt{I(2n)^{1/4}}$ and the bound reproduces the balanced $\sqrt{n}$ scaling with NO log at all, consistent with the balanced route (the logs price only genuine multi-scale amplitude spread); monotonicity — $\log^{3/2} < \log^2$ for all $n \geq 1$ ($\log_2(2n) \geq 1$), so the sharpening is an improvement everywhere, never a trade; dimension — the bound remains $\lambda^2 I^2 \times$ dimensionless count factors, as it must.

6. Devil's-advocate (three objections; CLAUDE.md 6.3)

- (α) “The Jensen step needs $J =$ the number of OCCUPIED classes; the proof's $J = \log_2(2n) + O(1)$ counts admissible indices. If occupied classes were fewer the original CS was already tight there.” DISMISSED in the favorable direction: Jensen with the occupied count $J_{\text{occ}} \leq J$ only IMPROVES the bound ($J_{\text{occ}}^{3/4} \leq J^{3/4}$); quoting J is the worst case. The sharpening is uniform.
- (β) “Minkowski-then-fourth-power may itself be the lossy step; perhaps a direct ℓ^2 assembly beats both versions and the half-log fight is moot.” VALID-with-mitigation: plausible — a direct assembly (orthogonality of transfer supports across well-separated class pairs, or a TT^* argument) is the natural next optimization and is registered; it would supersede, not contradict, this note. The half-log is what is PROVABLE today in two lines on the existing architecture.
- (γ) “The section-3 candidate ($8 \rightarrow 4$) is the kind of constant-fiddling that caused catch #7 (the 28/13 slip); flagging it at all risks premature use.” DISMISSED by the handling: the candidate

is explicitly NOT asserted, carries its decision criterion (the script's class-intensity convention), and is routed through the registered machine check before any use — exactly the post-catch-#7 discipline. The theorem of section 2 does not depend on it.

7. Result footer

Result ID:	B5-BEYOND-LAYER-BOUND / dyadic-lift-log-sharpening (unit U12)
Precise statement:	THEOREM (T3, proof fully written): the unconditional-amplitude lift improves to $\sum w_t^2 \leq 64 \sqrt{7} \lambda'^2 I^2 \sqrt{2n} \log^{\{3/2\}}(2n) + O(\lambda'^2 I^2)$ -- half a log removed by restoring $\sum_j N_j = 2n$ via CS + Jensen (x2.54 at $n = 44$; route n-reach x6.5). CANDIDATE further x4 (8 $\rightarrow$ 4 in the class cap) flagged conditional on a definition check; NOT asserted.
Scope:	Unconditional-amplitude route only; official ledger (balanced route) and amended-class closure unchanged.
Dependencies:	AddA v1.2 lift architecture (classes, per-class interpolation, bilinear CS, tail absorption -- all reused verbatim).
Evidence grade:	ANALYTIC (two textbook steps) at T3 pending the registered machine spot-check.
Reproduction command:	(registered) extend the AddA unbalanced-profile measurements to the sharpened ceiling + the section-3 definition check.
Expected output:	measured/ceiling ratio ~ 0.08 (two orders of slack preserved); definition-check verdict on the 8-vs-4; JSON under <code>claims/B5-BEYOND-LAYER-BOUND/runs/</code> .
Falsification gate:	A configuration violating the $\log^{\{3/2\}}$ ceiling (would refute the two-line argument -- not expected: both steps are inequalities with stated convexity directions); the definition check resolving 8-vs-4 either way is NOT a falsification (the candidate is conditional by construction).
Tier before / after:	No tier action; B5 stays T5-CANDIDATE.
No-overclaim statement:	The section-3 x4 is a flagged candidate, not a result. The sharpening does not change any closure margin (the amended class operates at $n \sim 44$ with orders of slack on every route). PDF build + linter PENDING operator-side.
Next required action:	Machine spot-check + definition check (registered); optional direct-l2-assembly

optimization (registered). Mainline continues
(U13).